Markov Chains

Assume k states of the world, which we label $1, 2, \ldots, k$. Let the probability that we move to state j from state i be p_{ij}, and call it the *transition probability*. Then the *transition probability matrix* of the Markov chain is the matrix

$$P = \begin{pmatrix} p_{11} & \cdots & p_{1k} \\ \vdots & \ddots & \vdots \\ p_{k1} & \cdots & p_{kk} \end{pmatrix}$$

For this matrix the rows are the current state of the world and the columns are the states into which we can move. Also the elements in the rows must add up to 1 or the system would have a positive probability of moving into an undefined state.

Markov Chains: Example

Say that there are three states of the world: rainy, overcast, and sunny. Let state 1 be rainy, 2 be cloudy, and 3 be sunny. Consider the transition matrix.

$$P = \begin{pmatrix} .8 & .1 & .1 \\ .3 & .2 & .5 \\ .2 & .6 & .2 \end{pmatrix}$$

For example, if it is rainy the probability that it remains rainy is .8. Also, if it is sunny, the probability that it becomes cloudy is .6.

Markov Chains: Example of Two Period Transition Matrix

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What if we want to know of a state in two periods? We can create a new transition matrix P_2, and $P_2 = PP$. Similarly it can be shown that $P_n = P^n$. For example, we can ask what is the probability that it will be cloudy the day after tomorrow?

$$P = \begin{pmatrix} .8 & .1 & .1 \\ .3 & .2 & .5 \\ .2 & .6 & .2 \end{pmatrix} \begin{pmatrix} .8 & .1 & .1 \\ .3 & .2 & .5 \\ .2 & .6 & .2 \end{pmatrix} = \begin{pmatrix} .69 & .16 & .15 \\ .40 & .37 & .23 \\ .38 & .26 & .36 \end{pmatrix}$$

Therefore, there is a 0.16 chance that if it is rainy today, then it will be cloudy the day after tomorrow.

Markov Chains: Probability Vector

Implicitly we have been assuming that we know with certainty the state of the world today. If we don't know the state of the world today we can define an initial vector of probabilities for today defined as $\mathbf{x}_0 = (p_1, p_2, \ldots, p_n)$. Then the probabilities of the state of the world tomorrow can be defined as $\mathbf{x}_0 P = \mathbf{x}_1$.
Generally, a vector $\mathbf{x}$ with $0 \le x_i \le 1$ for all i and $\sum_{i=1}^n x_i = 1$ is called a *probability vector*. For example,

$$\mathbf{x}_2 = \mathbf{x}_1 P = \mathbf{x}_0 P^2 = \begin{pmatrix} .3 & .6 & .1 \end{pmatrix} \begin{pmatrix} .8 & .1 & .1 \\ .3 & .2 & .5 \\ .2 & .6 & .2 \end{pmatrix} \begin{pmatrix} .8 & .1 & .1 \\ .3 & .2 & .5 \\ .2 & .6 & .2 \end{pmatrix}$$

$$= \begin{pmatrix} .485 & .296 & .219 \end{pmatrix}$$

There is a 0.296 chance that it will be cloudy day after tomorrow.

Steady State

A transition matrix P is called *regular* if P^n has only strictly positive entries for some integer n. A *steady state* vector $\mathbf{q}$ of a transition matrix P is a probability vector that satisfies the equation $\mathbf{q}P = \mathbf{q}$. Notice that this implies $P'\mathbf{q}' = \mathbf{q}'$ For an eigenvalue of $\lambda = 1$ this also implies $P'\mathbf{q}' = \lambda\mathbf{q}'$ or $(P' - \lambda I)\mathbf{q}' = \mathbf{0}$. If P is a regular transition matrix the following holds true:

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- 1 is an (left) eigenvalue of P or an eigenvalue of P'
- There is a unique probability vector $\mathbf{q}$, which is a (left) eigenvector associated with the eigenvalue 1 and $\mathbf{q}'$ is an eigenvector of P'.
- $\mathbf{x}P^n \to \mathbf{q}$ as $n \to \infty$ In the long-run this system converges to the steady state probability vector

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- A Markov Chain is Ergodic the chain has a unique steady state and converges to this steady state from all initial states.
  - A Markov Chain is Ergodic if the process is aperiodic and all states communicate.

Steady State: Example

What is the probability it will be cloudy as $n \to \infty$? Our transition probability matrix is

$$\begin{pmatrix} .2 & .8 \\ .5 & .5 \end{pmatrix}$$

The matrix P is regular since P only has strictly positive entries. Then to find the long-run probabilities we must find the vector $\mathbf{q}$. First we use the fact that when $\lambda = 1$ it is the case that $\mathbf{q}P = \lambda\mathbf{q} \implies P'\mathbf{q}' = \lambda\mathbf{q}'$. Therefore, $\mathbf{q}'$ is the eigenvector of $\mathbf{P}'$ that corresponds to the eigenvalue $\lambda = 1$. So we must find this eigenvector.

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We know that a steady state exists for this matrix since it is regular and since it is ergodic.We can solve for the steady state by solving the system $P'\mathbf{q}' = (1)\mathbf{q}'$. This system is

$$P = \begin{pmatrix} .2 & .5 \\ .8 & .5 \end{pmatrix} \begin{pmatrix} q_1 \\ q_2 \end{pmatrix} = \begin{pmatrix} q_1 \\ q_2 \end{pmatrix}$$

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Thus, $q_1 = \frac{5}{8}q_2$ and the eigenvector corresponding to the eigenvalue of 1 is

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The elements in this vector must sum to 1 so we divide the vector by the scalar $\frac{5}{8} + \frac{8}{8} = \frac{13}{8}$ or multiply the vector by $\frac{8}{13}$ to get

$$\begin{pmatrix} \frac{5}{13} \\ \frac{8}{13} \end{pmatrix}$$

Ergodic Failure

Consider the case, in which $\mathbf{x}_0 = \begin{pmatrix} 1 \\ 0 \end{pmatrix}$ and $P = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$
Then, chain just fluctuates between state 1 and state 2. So the states communicate but the chain is not aperiodic. It takes two periods to for a state to return. The unique steady state in this case is $\begin{pmatrix} .5 \\ .5 \end{pmatrix}$, but the chain only converges to this state if $\mathbf{x}_0 = \begin{pmatrix} .5 \\ .5 \end{pmatrix}$

Functions from $\mathbb{R}^n \to \mathbb{R}^m$

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- Finally a function can also take multiple inputs and multiple outputs, mapping $\mathbb{R}^n \to \mathbb{R}^m$. This is just an array of functions each mapping $\mathbb{R}^n \to \mathbb{R}$, $f(x_1, x_2, ..., x_n) = (f_1(x_1, x_2, ..., x_n), f_2(x_1, x_2, ..., x_n), ..., f_m(x_1, x_2, ..., x_n))$.

Visualizing functions that map $\mathbb{R}^n \to \mathbb{R}$

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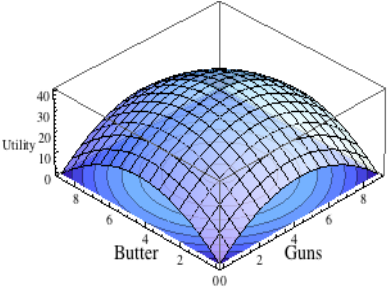

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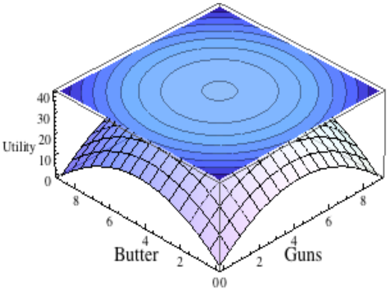

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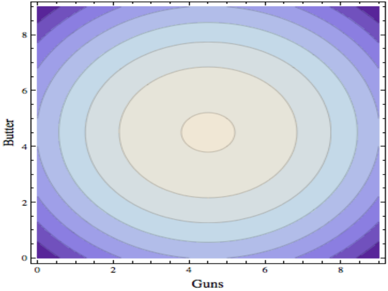

Partial Derivatives

Consider the function $f(x) = f(x_1, x_2, ...x_n)$. What is the partial derivative?

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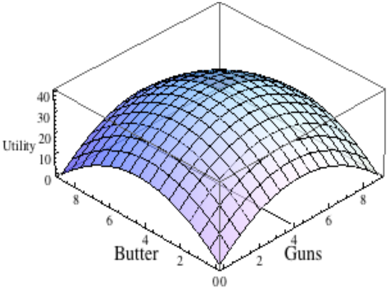
Consider the function $f(x) = f(x_1, x_2, ...x_n)$. What is the partial derivative? The partial derivative of f with respect to x_1, denoted $\frac{\partial f}{\partial x_1}$ or f_{x1}, is obtained by differentiating f with respect to x_1 and treating all of the other variables as constant:

$$f_{x1}(x_1, x_2, ..., x_n) = \lim_{h \to 0} \frac{f(x_1+h, x_2, ..., x_n) - f(x_1, x_2, ..., x_n)}{h}$$

Interpretation

$\frac{\partial}{\partial x1} f(x)$ is the change in the value of the function if x_1 changes infinitesimally. It is the slope of the line tangent to the graph of f at point $\mathbf{x}$ in the x_1 direction.



Example with Linear Regression

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- What is $\frac{\partial f(x_1,x_2,x_3)}{\partial x_2}$?

The Gradient

The gradient of a function f is a vector that contains all of the partial derivatives of a function:

$$Df(x_1, x_2, ..., x_n) = (\frac{\partial f}{\partial x_1}, \frac{\partial f}{\partial x_2}, ..., \frac{\partial f}{\partial x_n})$$

The transpose of this vector, evaluated at a specific point $a = (x_1^*, x_2^*, ..., x_n^*)$ is called the gradient of f at a or $\Delta f(a)$

Interpretation of the Gradient

The gradient is a list that changes in the value of the function if each variable separately changes infinitesimally. It gives the slope of the tangent lines to the graph of f at point x in each of the coordinate directions. The vector $\Delta f(a)$ also points into the direction in which f increases most rapidly.

Example of the Gradient

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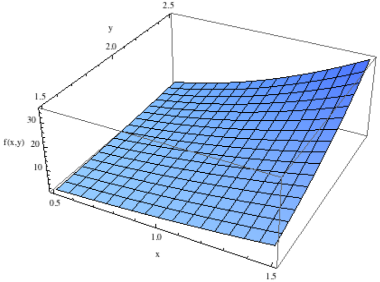
Example of Gradient

Consider the function $f(x,y) = x^2y^3$ and $a = (1,2)$.



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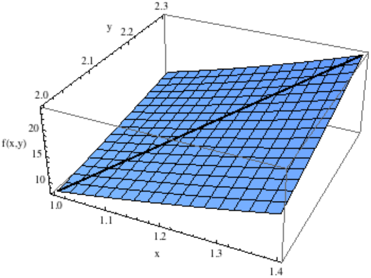
Consider the function $f(x, y) = x^2 y^3$ and $a = (1, 2)$. Then $\Delta f = \begin{pmatrix} 2xy^3 \\ 3x^2y^2 \end{pmatrix}$ and $\Delta f(a) = \begin{pmatrix} 16 \\ 12 \end{pmatrix}$.



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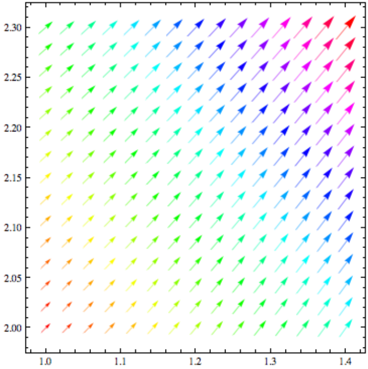

The Total Differential

The total differential of a function $f(x_1, x_2, ..., x_n)$ at a point $a = (x_1^*, x_2^*, ..., x_n^*)$ is

$$df = \frac{\partial f}{\partial x_1}(a)dx_1 + \frac{\partial f}{\partial x_2}(a)dx_2 + ... + \frac{\partial f}{\partial x_n}(a)dx_n$$

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Interpretation: Consider a tangent plane to the function $f(x, y)$. Let dx and dy be thought of as small deviations from the points x^* and y^*. The total differential states that the change in the value of the function f from this deviation from (x^*, y^*) to $(x^* + dx, y^* + dy)$ is the slope of the plane tangent to the function $f(x, y)$ at (x^*, y^*) in the x direction times the deviation dx, plus the slope of the plane at (x^*, y^*) in the y direction times the deviation dy.

The Total Differential: Example

Given the utility function of the form $U(C, N) = \ln(C) + \ln(N)$, find the approximate change in the utility by consuming an additional 0.1 units of C and less 0.1 units of N. Assume that the individual was originally consuming 0.5 units of each.

- What is $\frac{\partial U(C,N)}{\partial C}$?

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The full expression is then:

$$dU \approx \frac{1}{C}dC + \frac{1}{N}dN = \frac{1}{0.5}0.1 + \frac{1}{0.5}(-0.1) = 0$$

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Why is this just an approximation?

Second Order Partial Derivatives

A partial derivative of a partial derivative is a second order partial derivative. We denote the derivative with respect to x_i of the derivative with respect to x_j of the function f as $\frac{\partial^2 f}{\partial x_i \partial x_j}$. If $i = j$ then the partial derivative is denoted as $\frac{\partial^2 f}{\partial x_i^2}$, if $i \neq J$ then $\frac{\partial^2 f}{\partial x_i \partial x_j}$ is also called the cross derivative. Note that $\frac{\partial^2 f}{\partial x_i \partial x_j} = \frac{\partial^2 f}{\partial x_i \partial x_j}$ if the cross derivatives exist and are continuous.

The Hessian Matrix

The Hessian Matrix is a matrix of the second order partial derivatives of a function. Consider a function $f(x_1, ..., f_n)$. Its Hessian is given by.

$$\begin{pmatrix} \frac{\partial^2 f}{\partial x_1^2} & \frac{\partial^2 f}{\partial x_2 x_1} & \cdots & \frac{\partial^2 f}{\partial x_n x_1} \\ \frac{\partial^2 f}{\partial x_1 x_2} & \frac{\partial^2 f}{\partial x_2^2} & \cdots & \frac{\partial^2 f}{\partial x_n x_2} \\ \vdots & \vdots & \ddots & \vdots \\ \frac{\partial^2 f}{\partial x_1 x_n} & \frac{\partial^2 f}{\partial x_2 x_n} & \cdots & \frac{\partial^2 f}{\partial x_n^2} \end{pmatrix}$$

A Hessian matrix is always a square matrix if $f \in C^2$ by Young's theorem (p. 330 in SB)

The Hessian Matrix: Example

Let $f(x, y) = x^2 y^3$. Then the Hessian is

$$\begin{pmatrix} 2y^3 & 6xy^2 \\ 6xy^2 & 6x^2 y \end{pmatrix}$$

The Total Derivative

Let $\mathbf{x}(t) = (x(t), ..., x_n(t))$. Also, let f be some function of $\mathbf{x}(t)$ and t. Then

$$\frac{df}{dt}(t, \mathbf{x}(t)) = \frac{\partial f}{\partial t} + \frac{\partial f}{\partial x_1} \cdot \frac{dx_1}{dt} + ... + \frac{\partial f}{\partial x_n} \cdot \frac{dx_n}{dt}$$

Directional Derivatives

To calculate a directional derivative we need a function $f(\mathbf{x})$ where $\mathbf{x} = (x_1, ..., x_n) \in \mathbb{R}^n$, a vector $\mathbf{v} = (v_1, ..., v_n)$, and a parameter t.

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$\frac{dg}{dt} = \frac{\partial f}{\partial x_1}(x_1+t\cdot v_1, ..., x_n+t\cdot v_n)\cdot v_1 + ... + \frac{\partial f}{\partial x_n}(x_1+t\cdot v_1, ..., x_n+t\cdot v_n)\cdot v_n$

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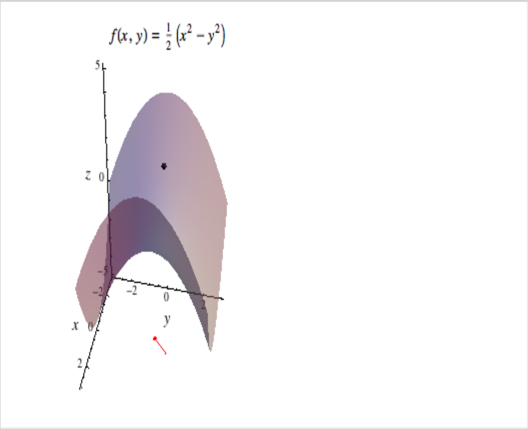
$$\frac{dg}{dt} = \frac{\partial f}{\partial x_1}(x_1+t\cdot v_1, ..., x_n+t\cdot v_n)\cdot v_1+...+\frac{\partial f}{\partial x_n}(x_1+t\cdot v_1, ..., x_n+t\cdot v_n)\cdot v_n$$

Since we are interested at f at the original point $\mathbf{x}$, we let $t = 0$ to get

$$\frac{dg}{dt}(0) = \frac{\partial f}{\partial x_1}(x_1, ..., x_n) \cdot v_1 + ... + \frac{\partial f}{\delta x_n}(x_1, ..., x_n) \cdot v_n = [\Delta f(x)]' \cdot \mathbf{v}$$

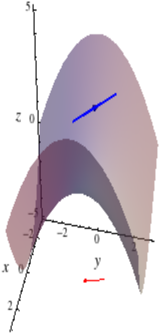
This is the dot product of the total derivative at $\mathbf{x}$ with the vector $\mathbf{v}$. This is also denoted as $Df_{\mathbf{x}} \cdot \mathbf{v}$ or $\frac{\partial f}{\partial \mathbf{v}}(\mathbf{x})$.

Visualizing Directional Derivatives



Visualizing Directional Derivatives

$$f(x, y) = \frac{1}{2}\left(x^2 - y^2\right)$$



$D_u f(x, y) = \nabla f(x, y) \cdot u$

$\nabla f(x, y) = (f_x, f_y)$

$f_x(x, y) = x$

$f_x(-2., 0.) = -2$

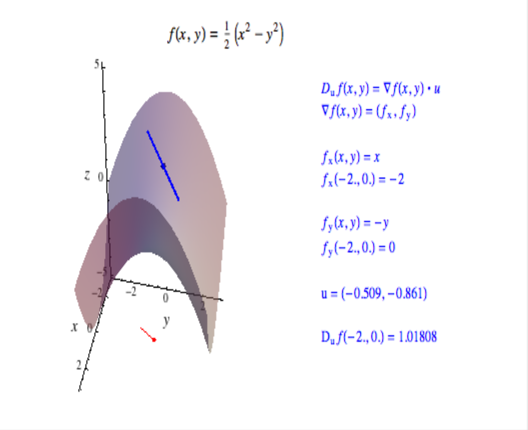
$f_y(x, y) = -y$

$f_y(-2., 0.) = 0$

$u = (-0.279, 0.96)$

$D_u f(-2., 0.) = 0.557982$

Visualizing Directional Derivatives



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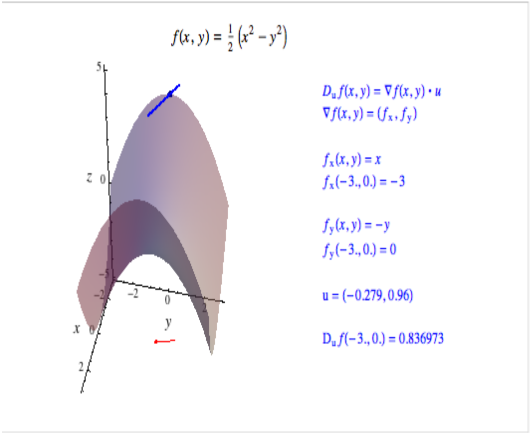

Derivatives of Multiple Functions of Several Variables

Consider m equations which are each a function of n variables:

$$y_1 = f_1(x_1, ..., x_n)$$
$$.$$
$$.$$
$$.$$
$$y_m = f_m(x_1, ..., x_n)$$

Derivatives of Multiple Functions of Several Variables

A partial derivative of this function is now defined as:

$$\begin{pmatrix} \frac{\partial f_1}{\partial x_1} \\ \vdots \\ \frac{\partial f_m}{\partial x_1} \end{pmatrix}$$

Derivatives of Multiple Functions of Several Variables: Jacobian

The Jacobian matrix, denoted Df, is an $n \times m$ matrix

$$\begin{pmatrix} \frac{\partial f_1}{\partial x_1}(\mathbf{x}*) & \dots & \frac{\partial f_1}{\partial x_n}(\mathbf{x}*) \\ \vdots & \ddots & \vdots \\ \frac{\partial f_m}{\partial x_1}(\mathbf{x}*) & \dots & \frac{\partial f_m}{\partial x_n}(\mathbf{x}*) \end{pmatrix}$$

Jacobian matrix: Example

Find the Jacobian of the following system:

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so the Jacobian is

$$\begin{pmatrix} \frac{1}{Y} & -\frac{X}{Y^2} \\ Y & X \end{pmatrix}$$

Concavity and Convexity

- A function is concave for a subset U of $\mathbb{R}^n$ if for all $\mathbf{x}$ and $\mathbf{y}$ in U and for all $\lambda \in [0, 1]$ it is the case that $f(\lambda\mathbf{x} + (1-\lambda)\mathbf{y}) \geq \lambda f(\mathbf{x}) + (1-\lambda)f(\mathbf{y})$.

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- If the Hessian, H, is either negative or positive definite at all points $\mathbf{x}$, then it is strictly concave or convex.
  - A matrix is negative definite iff its eigenvalues are strictly negative. A matrix is positive definite iff its eigenvalues are strictly positive.

Integration over Several Variables

Consider $f : \mathbb{R}^2 \to \mathbb{R}$. How is the following integral computed?

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We compute the integral by treating $f(x, y)$ as a function of x, and treating y as a constant. Note that the result of the integration will still be a function of y.

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Evaluate $\int_2^4 (x+y)\,dx$

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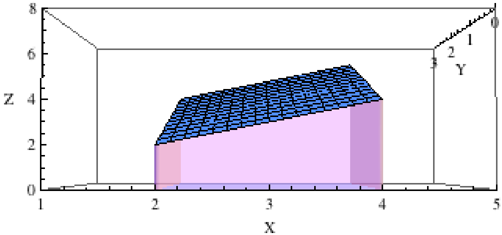
$$\int_2^4 (x+y)\,dx = \frac{1}{2}(x^2 + xy)|_2^4 = 6 + 2y$$



Double Integrals

To evaluate the double integral

$$\int_a^b \int_c^d f(x, y)\,dxdy$$

it is helpful to rewrite the integral as

$$\int_a^b (\int_c^d f(x, y)\,dx)\,dy$$

and evaluate the inside integral first. Then the outside integral can be evaluated to obtain the solution.

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Double Integrals: Example

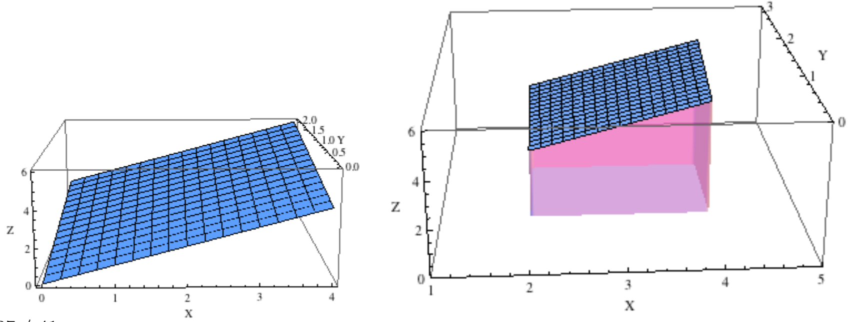
Evaluate $\int_1^2 \int_2^4 (x+y)\,dxdy$

$$\int_1^2 \int_2^4 (x+y)\,dxdy = \int_1^2 (\int_2^4 (x+y)\,dx)\,dy$$
$$= \int_1^2 ((\tfrac{1}{2}x^2 + xy)|_2^4)\,dy = \int_1^2 (6+2y)\,dy$$

Double Integrals: Example

Evaluate $\int_1^2 \int_2^4 (x+y)\,dxdy$

$$\begin{aligned}\int_1^2 \int_2^4 (x+y)\,dxdy &= \int_1^2 \left(\int_2^4 (x+y)\,dx\right)dy \\ &= \int_1^2 \left(\left(\tfrac{1}{2}x^2 + xy\right)\Big|_2^4\right)dy = \int_1^2 (6+2y)\,dy \\ &= 6y + y^2\Big|_1^2 = 9\end{aligned}$$



Higher Order Integrals

Triple and higher order integrals are evaluated in the same manner as doubles; first integrate with respect to one variable and work back towards lower order integrals

$$\int_a^b \int_c^d \int_e^f f(x,y,z)\,dxdydz = \int_a^b (\int_c^d (\int_e^f f(x,y,z)\,dx)\,dy)\,dz$$

Order of Integration

When integrating over a rectangle (or its equivalent in higher dimensions), the order of the integration does not matter.

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When integrating over a rectangle (or its equivalent in higher dimensions), the order of the integration does not matter. For example, recall the double integral $\int_1^2 \int_2^4 (x+y)\,dxdy$, which we integrated with respect to x first. We can simply change the order of integration and derive the same answer. (Note that we are integrating over a rectangle.)

$$\int_2^4 \int_1^2 (x+y)\,dydx = \int_2^4 (\int_1^2 (x+y)\,dy)\,dx = \int_2^4 ((\tfrac{1}{2}y^2 + xy)|_1^2)\,dx$$
$$\int_2^4 (\tfrac{3}{2} + x)\,dy = \tfrac{3}{2}x + \tfrac{1}{2}x^2|_2^4 = 3 + 6 = 9$$

Order of Integration

When we are integrating to variables that are also limits then the order of integration matters. We need to integrate over the all of the variables that are not in the limit first.

$$\int_2^4 \int_1^x (x+y)\,dydx = \int_2^4 (\int_1^x (x+y)\,dy)\,dx = \int_2^4 ((\tfrac{1}{2}y^2 + xy)|_1^x)\,dx$$
$$\int_2^4 (\tfrac{3}{2}x^2 - x - \tfrac{1}{2})\,dy = \tfrac{1}{2}x^3 - \tfrac{1}{2}x^2 - \tfrac{1}{2}x|_2^4 = 32 - 8 - 2 = 22$$